Reproduction Guide

This guide provides step-by-step instructions to reproduce Gravisonde locally and on AWS.

---

1. Prerequisites

- **OS:** Linux, macOS, or Windows with WSL2
- **Docker:** >= 24.0
- **Node.js:** >= 22 (for dashboard)
- **Python:** >= 3.10 (for pipeline)
- **AWS CLI:** configured with credentials
- **AWS CDK:** v2.x installed (`npm i -g aws-cdk`)

---

2. Local Development

2.1 Clone & Navigate

```
git clone <repo-url>
cd gravisonde
```

2.2 Create Environment File

```
cp .env.example .env
# Edit .env if needed (LEADERBOARD_ID for leaderboard API)
```

2.3 Start Local Stack

```
make local-up
```

This runs:

Service	Port	Purpose
dashboard	3000	Next.js app with SSE metrics
pipeline	N/A	Stub OpenCV 5 pipeline (simulated)
localstack	4566	Mock AWS (CloudWatch, S3)

2.4 Verify

Open http://localhost:3000 in your browser. You should see the dashboard with:

- Live metrics cards (FPS, latency, cost)
- Charts updating in real-time
- Download button for benchmark CSV

2.5 Stop Local Stack

```
make local-down
```

---

3. AWS Deployment

3.1 Bootstrap CDK (once per account/region)

```
cd infra
make bootstrap
```

This creates the CDK bootstrapped resources (bucket, lambda, etc.).

3.2 Deploy Infrastructure

```
make deploy-infra
```

CDK outputs:

- `InstanceId` — EC2 instance ID
- `InstancePublicIp` — Public IP of the pipeline server
- `BenchmarkBucketName` — S3 bucket for CSV artifacts
- `CwLogGroup` — CloudWatch log group

3.3 Build & Push Pipeline Images

```
make build-pipeline
make push-pipeline
```

This builds:

- `cv-pipeline:cool` — COOL runtime image (for EC2)
- `cv-pipeline:stock` — Stock OpenCV 5 image (for comparison)

3.4 Start Pipeline on EC2

```
make start-pipeline
```

The pipeline starts via AWS Systems Manager (SSM) and begins streaming metrics to CloudWatch and S3.

3.5 Run Benchmark

```
make benchmark
```

Runs COOL vs Stock for a fixed duration and uploads CSV to the S3 bucket.

---

4. Verification Checklist

Step	How to Verify
Dashboard loads	http://localhost:3000 or https://gravisonde.demo (https://gravisonde.demo)
SSE stream active	Network tab → ws://localhost:3000/api/metrics
Pipeline running	Check EC2 instance status in AWS console
Metrics in CloudWatch	CW namespace `Gravisonde`

Step	How to Verify
CSV in S3	`aws s3 ls s3://gravisonde-benchmarks-<id>/`

---

5. Troubleshooting

Symptom	Fix
Dashboard shows "No data"	Ensure pipeline is running (`make start-pipeline`)
LocalStack connection refused	`make local-down && make local-up`
Python import errors	`pip install -r apps/cv-pipeline/requirements.txt`
TypeScript errors	`cd apps/dashboard && npm install && npx tsc --noEmit`
CDK synth fails	Check `coolAmiId` parameter is set in `bin/gravisonde.ts`

---

6. Clean Up

```
make destroy-infra   # Remove CDK stack
make local-down      # Stop Docker compose
```

---

7. Re-create Sample Video

The sample video is hosted in `docker-compose.yml` volume mount:

```
volumes:
  - ./sample_video.mp4:/data/sample.mp4:ro
```

If the file is missing or 0 bytes, generate a test video:

```
ffmpeg -f lavfi -i testsrc=duration=30:size=640x480:rate=30 -c:v libx264 -preset fast output.mp4
mv output.mp4 sample_video.mp4
```

Or download a CC0 test video:

```
wget https://commondatastorage.googleapis.com/gtv-videos-bucket/sample/ForBiggerBlazes.mp4 -O sample_video.mp4
```